CODE

main.py | model.py | preprocess.py | similarity.py | suggestions.py | vectorizer.py

main.py

```python
import streamlit as st
from model import analyze_resume

st.title("📄 Smart Resume Analyzer (SkillLens)")

resume = st.text_area("Paste your Resume")
job_desc = st.text_area("Paste Job Description")

def get_grade(score):
    if score >= 80:
        return "A (Excellent)"
    elif score >= 60:
        return "B (Good)"
    elif score >= 40:
        return "C (Average)"
    else:
        return "D (Needs Improvement)"

if st.button("Analyze"):
    if resume and job_desc:
        score, suggestions = analyze_resume(resume, job_desc)

        percentage = round(score * 100, 2)
        grade = get_grade(percentage)

        # ✅ Final Score
        st.success(f"✅ Resume Score: {percentage}%")
        st.info(f"📊 Grade: {grade}")

        # ✅ Progress bar
        st.progress(int(percentage))

        st.subheader("💡 Suggestions:")
        for s in suggestions:
            st.write("- " + s)
    else:
        st.warning("Please enter both resume and job description")
```

main.py | model.py | preprocess.py | similarity.py | suggestions.py | vectorizer.py

model.py

```python
from preprocess import clean_text
from vectorizer import get_vectors
from similarity import calculate_similarity
from suggestions import give_suggestions

def analyze_resume(resume, job_desc):
    clean_resume = clean_text(resume)
    clean_job = clean_text(job_desc)

    vectors = get_vectors([clean_resume, clean_job])
    score = calculate_similarity(vectors)

    suggestions = give_suggestions(resume, job_desc)

    return score, suggestions
```

main.py | model.py | preprocess.py | similarity.py | suggestions.py | vectorizer.py

preprocess.py

```python
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize

# ✅ Fix (add this)
nltk.download('punkt')
nltk.download('punkt_tab')   # <-- ye line ADD karo
nltk.download('stopwords')

def clean_text(text):
    words = word_tokenize(text.lower())
    filtered = [w for w in words if w.isalnum() and w not in stopwords.words('english')]
    return " ".join(filtered)
```

main.py | model.py | preprocess.py | similarity.py | suggestions.py | vectorizer.py

similarity.py

```python
from sklearn.metrics.pairwise import cosine_similarity

def calculate_similarity(vectors):
    return cosine_similarity(vectors[0:1], vectors[1:2])[0][0]
```

main.py | model.py | preprocess.py | similarity.py | suggestions.py | vectorizer.py

suggestions.py

```python
def give_suggestions(resume, job_desc):
    suggestions = []

    resume = resume.lower()
    job_words = set(job_desc.lower().split())

    missing = [word for word in job_words if word not in resume]

    for word in list(missing)[:5]:
        suggestions.append(f"Consider adding '{word}'")

    if "project" not in resume:
        suggestions.append("Add project section")

    if "skill" not in resume:
        suggestions.append("Add skills section")

    if len(resume.split()) < 80:
        suggestions.append("Increase resume content")

    return suggestions
```

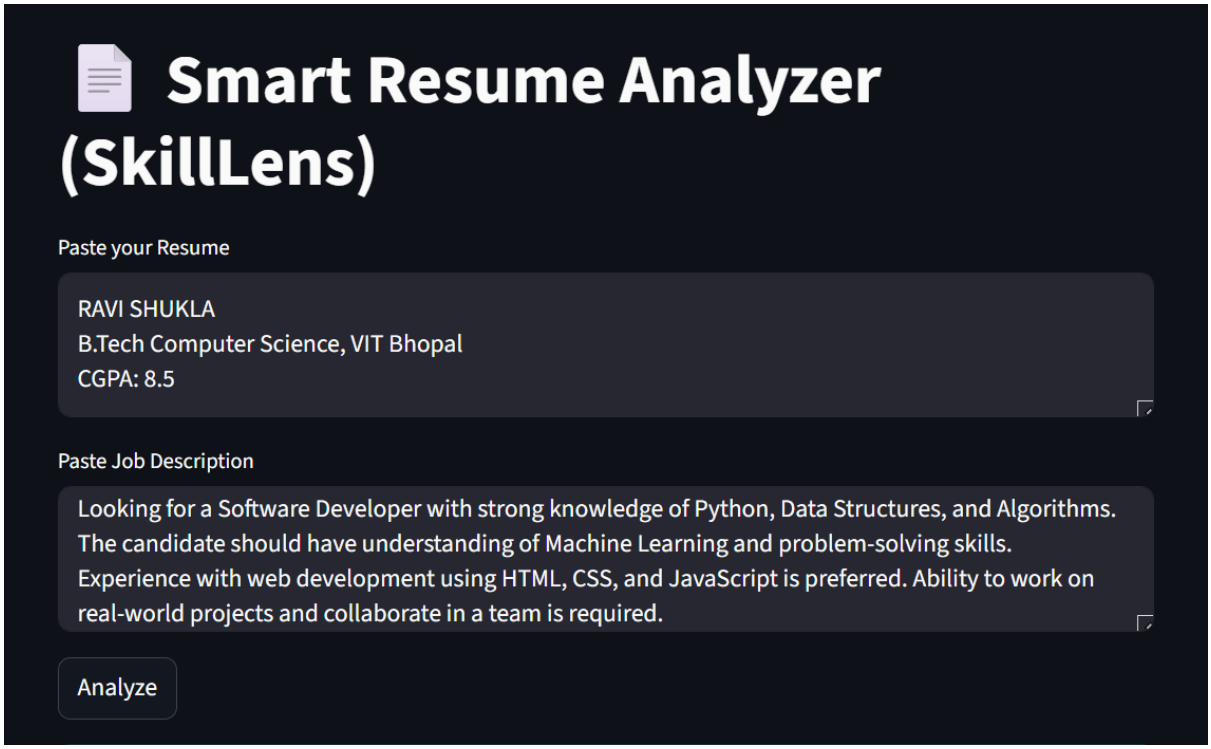
main.py | model.py | preprocess.py | similarity.py | suggestions.py | vectorizer.py

vectorizer.py

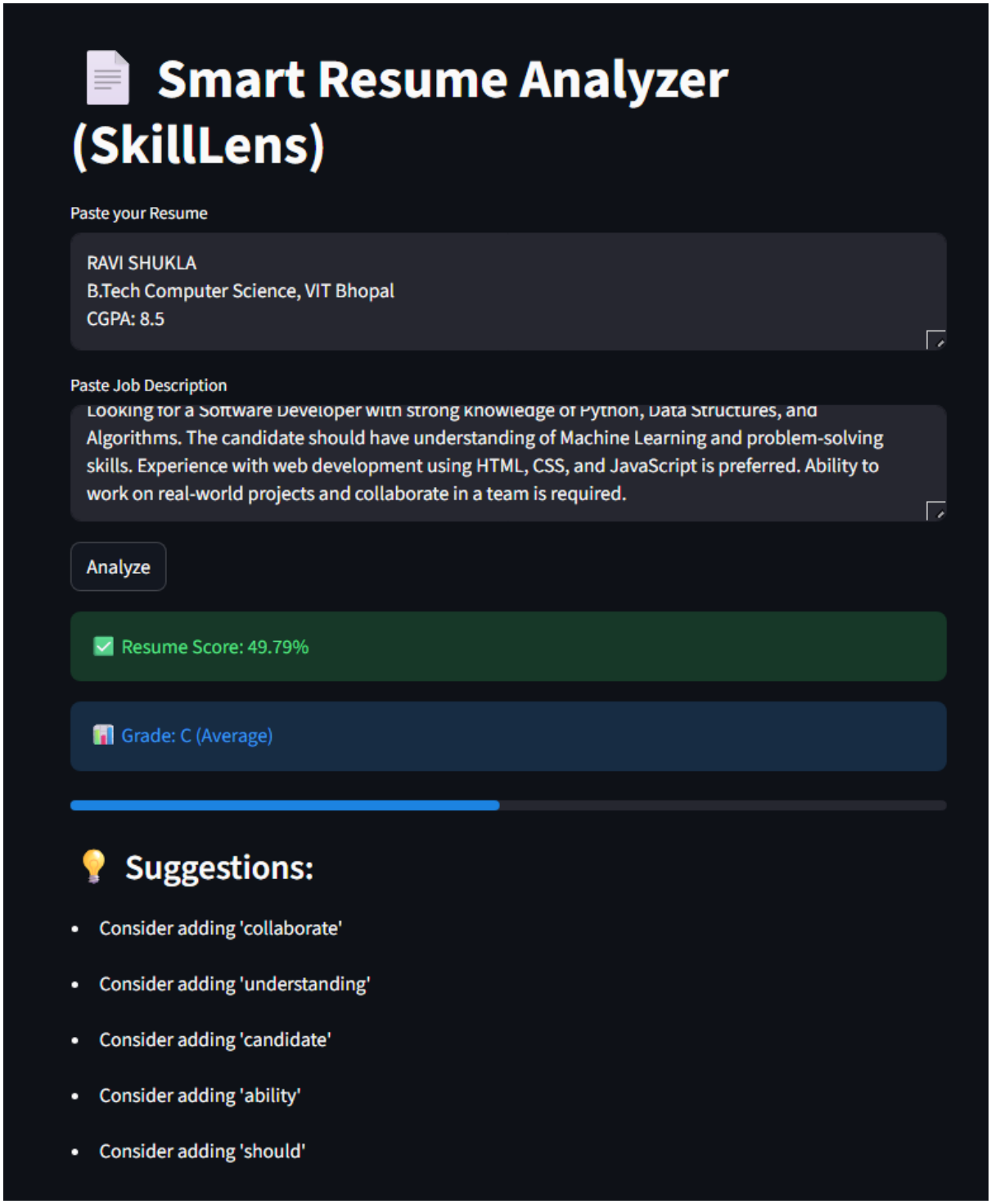
```python
from sklearn.feature_extraction.text import TfidfVectorizer

def get_vectors(texts):
    tfidf = TfidfVectorizer()
    return tfidf.fit_transform(texts)
```

INPUT


Smart Resume Analyzer (SkillLens)
Paste your Resume
RAVI SHUKLA
B.Tech Computer Science, VIT Bhopal
CGPA: 8.5
Paste Job Description
Looking for a Software Developer with strong knowledge of Python, Data Structures, and Algorithms. The candidate should have understanding of Machine Learning and problem-solving skills. Experience with web development using HTML, CSS, and JavaScript is preferred. Ability to work on real-world projects and collaborate in a team is required.
Analyze

OUTPUT


Smart Resume Analyzer (SkillLens)
Paste your Resume
RAVI SHUKLA
B.Tech Computer Science, VIT Bhopal
CGPA: 8.5
Paste Job Description
Looking for a Software Developer with strong knowledge of Python, Data Structures, and Algorithms. The candidate should have understanding of Machine Learning and problem-solving skills. Experience with web development using HTML, CSS, and JavaScript is preferred. Ability to work on real-world projects and collaborate in a team is required.
Analyze
Resume Score: 49.79%
Grade: C (Average)
Suggestions:
Consider adding 'collaborate'
Consider adding 'understanding'
Consider adding 'candidate'
Consider adding 'ability'
Consider adding 'should'